



Summer Internship Report

on

OmniServe: Wi-Fi Controlled Omni-Directional Robot with Encoder Feedback and Pick-and-Place Arm

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Certificate

This is to certify that the internship project entitled **“OmniServe: Wi-Fi Controlled Omni-Directional Robot with Encoder Feedback and Pick-and-Place Arm”** was successfully completed by the following students in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (ECE) under the internship program at AICTE IDEA Lab-Guru Gobind Singh Indraprastha University, New Delhi - 110078.

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Declaration

We hereby declare that the project work presented in this internship report entitled **“OmniServe: Wi-Fi Controlled Omni-Directional Robot with Encoder Feedback and Pick-and-Place Arm”** is entirely our own work and has not been submitted for any degree or diploma from this or any other institute for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (ECE).

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Abstract

This project presents **OmniServe**, a Wi-Fi-controlled, omni-directional robotic platform designed for enhanced mobility and object manipulation in constrained environments. The robot features precise movement capabilities using encoder feedback and a 3-DOF robotic arm for pick-and-place operations. It can move in all eight directions and rotate in place, enabling agile navigation. A web-based dashboard facilitates real-time control, system monitoring, PID tuning, and vector-based motion commands. Using a PID algorithm, the robot can perform accurate linear movements with control over both distance and direction. With its integrated robotic arm, OmniServe extends its utility to applications such as warehouse automation, smart delivery systems, and educational robotics.

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Introduction

Motivation

The rapid advancement of automation and robotics has revolutionized industries, logistics, and daily life. However, the need for versatile, precise, and remotely controllable robots remains a significant challenge, especially in constrained environments such as warehouses and smart factories. OmniServe was conceived to address these challenges by providing a robust, omni-directional robotic platform with advanced feedback and manipulation capabilities.

Problem Statement

Traditional mobile robots are often limited by their drive mechanisms, restricting their movement to forward, backward, and simple turns. This limitation hinders their effectiveness in environments requiring agile navigation and precise positioning. Additionally, the lack of real-time feedback and manipulation capabilities further restricts their utility in complex tasks such as pick-and-place operations.

Objectives

The primary objectives of the OmniServe project are:

- To design and develop a Wi-Fi controlled omni-directional robot capable of 360-degree movement.
- To implement encoder feedback for precise distance and direction control.
- To integrate a 3-DOF robotic arm for pick-and-place operations.
- To develop a web-based dashboard for real-time control, monitoring, and PID tuning.
- To demonstrate the robot's applicability in warehouse automation, smart delivery, and educational robotics.

Report Structure

This report is organized as follows:

- **Literature Survey:** Reviews related work and existing technologies in omni-directional robotics and feedback systems.
- **System Overview:** Provides a high-level description of the hardware and software components.
- **Hardware and Software Architecture:** Details the design and implementation of the robot's physical and control systems.
- **Methodology:** Describes the development process, including design, prototyping, and testing.
- **Features and Applications:** Highlights the key features and potential use cases of OmniServe.
- **Results and Discussion:** Presents experimental results, performance analysis, and insights gained.
- **Challenges Faced:** Discusses obstacles encountered and solutions implemented.
- **Future Work:** Outlines possible enhancements and research directions.
- **Conclusion:** Summarizes the project outcomes and its significance.

Literature Survey

Omni-Directional Robots

Omni-directional robots have been widely researched for their ability to move in any direction without changing orientation, making them ideal for applications in constrained environments. Mecanum and omni-wheel designs are commonly used to achieve this mobility [1, 2].

Encoder Feedback and PID Control

Precise movement in mobile robots is often achieved using quadrature encoders and PID (Proportional-Integral-Derivative) control algorithms. Encoder feedback allows for accurate distance and speed measurement, while PID controllers help maintain desired trajectories [3, 4].

Robotic Arms and Manipulation

Robotic arms with multiple degrees of freedom (DOF) are essential for pick-and-place tasks. The integration of servo motors and inverse kinematics enables precise object manipulation [5, 6].

Web-Based Control Systems

Recent advancements in IoT and embedded systems have enabled real-time robot control via web interfaces. Technologies such as WebSockets and REST APIs facilitate seamless communication between users and robots [7, 8].

Summary

The literature indicates that combining omni-directional mobility, encoder feedback, PID control, and web-based interfaces can significantly enhance the versatility and usability of mobile robots in industrial and educational settings.

System Overview

System Architecture

OmniServe is designed as a modular, Wi-Fi controlled omni-directional robot with encoder feedback and a 3-DOF robotic arm. The system is divided into several subsystems: drive, feedback, control, and manipulation.

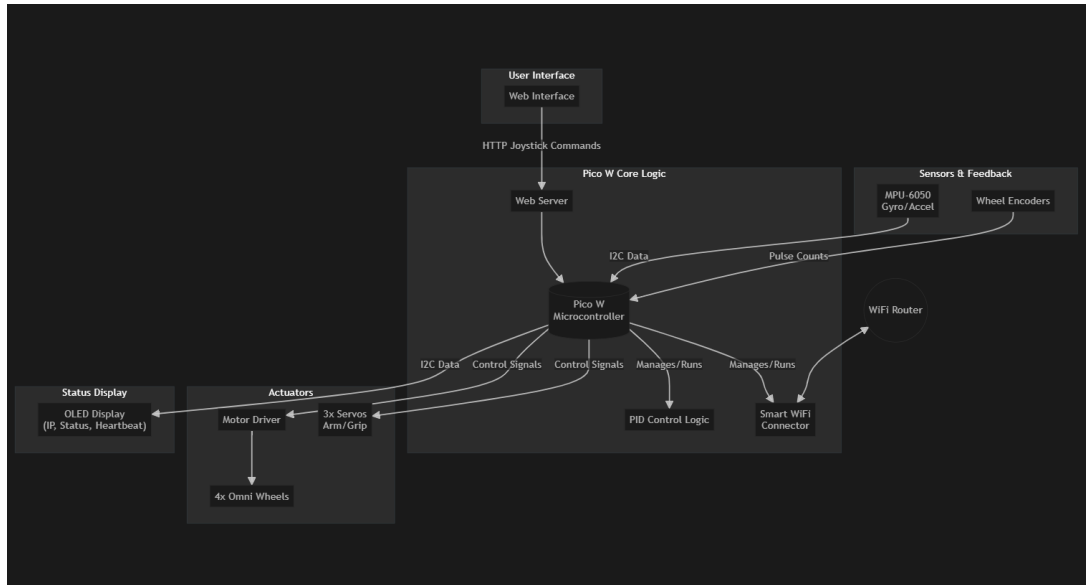


Figure 1: Block diagram of the OmniServe system.

Main Components

- **Controller:** Raspberry Pi Pico W for main control and Wi-Fi communication.
- **Motors:** Four DC motors with omni-wheels for 360-degree movement.
- **Motor Drivers:** Dual TB6612FNG drivers for efficient motor control.
- **Encoders:** Quadrature encoders on each wheel for feedback.
- **Robotic Arm:** 3-DOF servo-based arm for pick-and-place tasks.
- **Web Dashboard:** Real-time control and monitoring interface.
- **Power Supply:** Battery pack with voltage regulation.

Module Summary

Module	Description
Controller	Raspberry Pi Pico W, runs main logic and hosts web dashboard
Motor Drivers	TB6612FNG, controls four DC motors
Encoders	Quadrature, provides feedback for PID control
Robotic Arm	3-DOF, servo-based, for object manipulation
Web Dashboard	HTML/JS interface for control and monitoring
Power Supply	Li-ion battery, voltage regulation for stable operation

Interaction

The controller receives commands from the web dashboard, processes encoder feedback, and drives the motors accordingly. The robotic arm is controlled via PWM signals, and all telemetry is sent back to the dashboard for real-time monitoring.

Hardware Architecture

Chassis and Mechanical Design

The robot uses a four-wheel omni-wheel chassis, enabling movement in any direction. The frame is constructed from lightweight aluminum for strength and portability. The omni-wheels (1.5-inch diameter) allow for smooth lateral and diagonal movement.

Motors and Drivers

Four DC motors are used, each controlled by a TB6612FNG dual motor driver. The drivers are chosen for their efficiency, compactness, and ability to handle the required current.

Encoders

Each wheel is equipped with a quadrature encoder (520 ticks/rev) for precise feedback. The encoders are essential for closed-loop PID control and accurate distance estimation.

Robotic Arm

A 3-DOF servo-based arm is mounted on the chassis. The arm uses standard hobby servos for base rotation, shoulder, and gripper movement, allowing for basic pick-and-place operations.

Power Supply

A rechargeable Li-ion battery pack powers the system, with voltage regulation to ensure stable operation for both logic and motors.

Wiring and Integration

All components are connected to the Raspberry Pi Pico W. Careful wiring and cable management ensure reliability and ease of maintenance.

GPIO Mapping

Function	Pico W Pin(s)
Motor Control	GP2–GP15
Encoder Inputs	GP16–GP22, GP26
Standby	GP8
Robotic Arm PWM	GP23, GP24, GP25 (suggested)
I2C (reserved)	GP0 (SDA), GP1 (SCL)

Hardware Photo

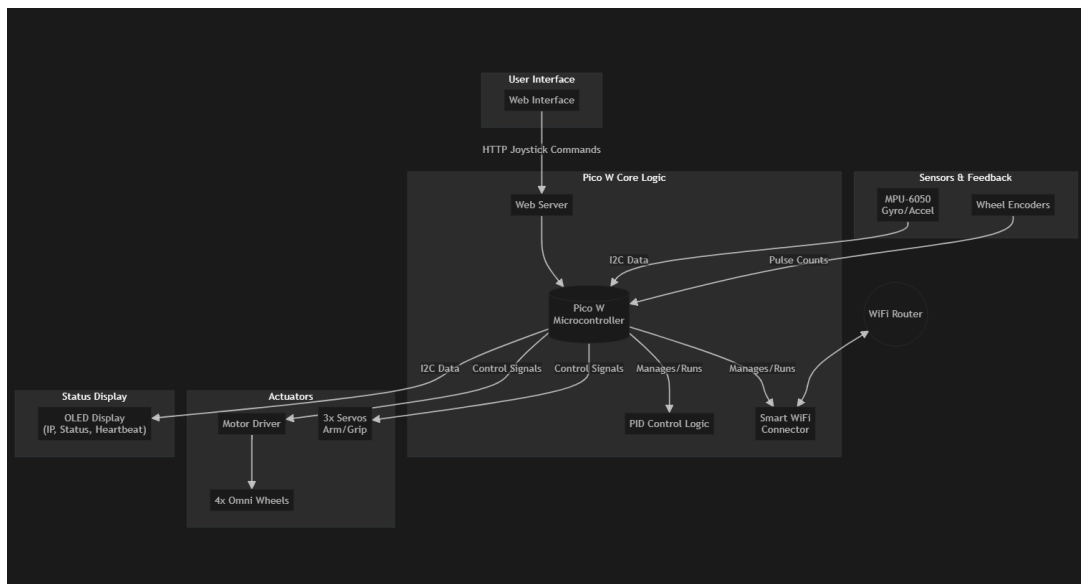


Figure 2: OmniServe hardware prototype.

Design Choices and Trade-offs

The use of omni-wheels provides maximum maneuverability but requires careful control logic. The Raspberry Pi Pico W was selected for its Wi-Fi capability and GPIO count. Power management was a key consideration to ensure long operational time and safe charging.

Software Architecture

Embedded Firmware

The core logic is implemented in Arduino C++ and runs on the Raspberry Pi Pico W. The firmware handles motor control, encoder feedback, PID computation, and communication with the web dashboard.

Web Dashboard

A responsive web dashboard (HTML, CSS, JavaScript) is served from the Pico W. It provides real-time control, telemetry, and PID tuning. Joystick input is handled using NippleJS, and all commands are sent via WebSockets.

Communication Protocols

- **WebSocket:** For low-latency, real-time control and telemetry.
- **HTTP/REST:** For configuration and status queries.

Control Logic

The robot supports both direct (WASD) and vector-based movement. Encoder feedback is used to close the loop for precise distance and direction control. The PID controller is tuned in real time from the dashboard.

PID Implementation

Listing 1: PID Control Loop

```
float error = target - measured;
integral += error * dt;
derivative = (error - prev_error) / dt;
output = Kp * error + Ki * integral + Kd * derivative;
prev_error = error;
```

Encoder Handling

Interrupts are used to read quadrature encoder signals, ensuring accurate tick counting even at high speeds.

Safety Features

An emergency stop is implemented in both firmware and dashboard. All commands are validated for safety before execution.

Software Flowchart

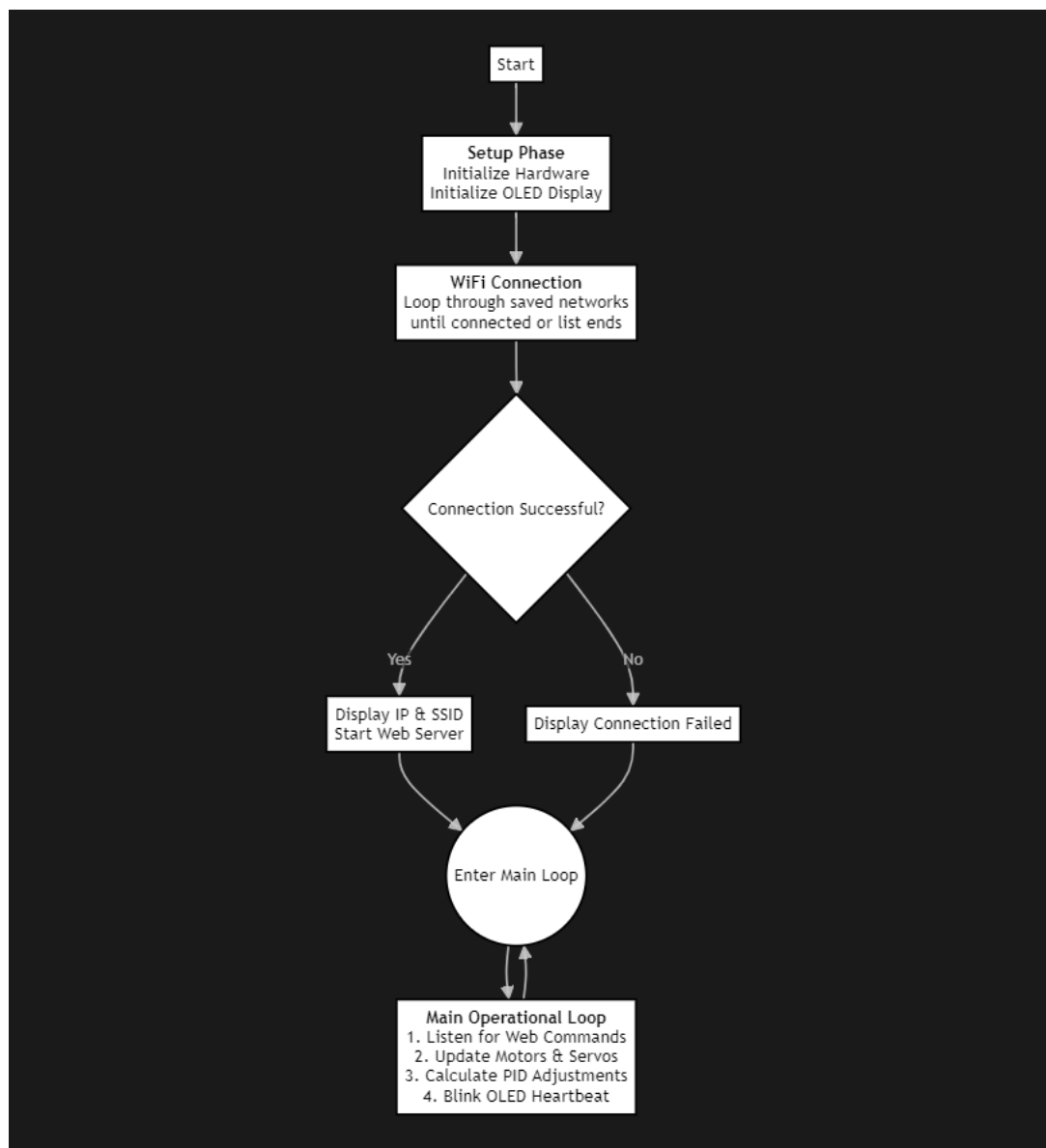


Figure 3: Software architecture flowchart.

Methodology

Design Phase

The project began with requirements analysis and system design. Block diagrams and schematics were created to plan the hardware and software integration.

Prototyping

Initial prototypes were built using breadboards and 3D-printed parts. Motor drivers, encoders, and the robotic arm were tested individually before full integration.

Integration

All subsystems were integrated on the main chassis. Careful attention was paid to wiring, power distribution, and signal integrity. The web dashboard was connected to the firmware for real-time control.

Testing and Validation

The robot was tested in various scenarios: straight-line travel, diagonal movement, rotation, and pick-and-place tasks. Encoder feedback and PID control were validated for accuracy. The dashboard was tested for responsiveness and reliability.

Iterative Development

Development followed an iterative approach, with regular testing and refinement. Issues were logged and addressed in each cycle.

Gantt Chart

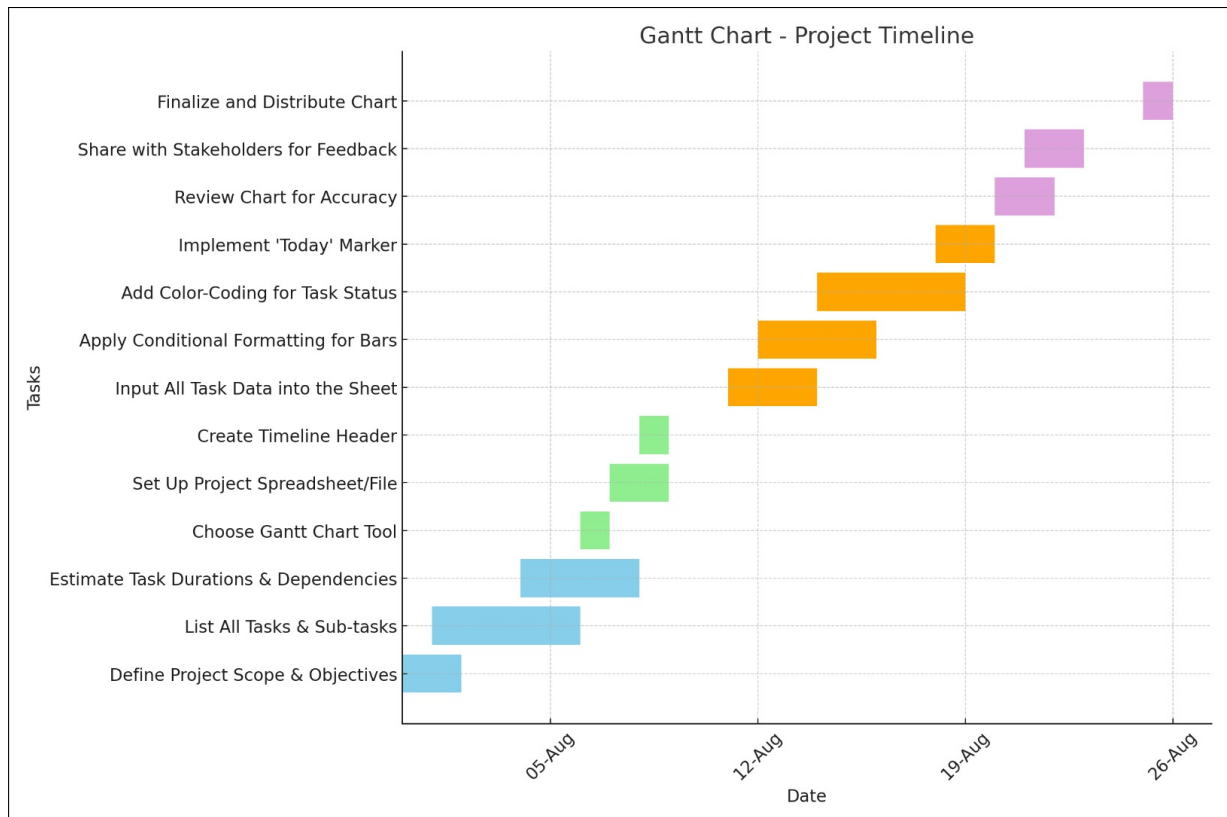


Figure 4: Project Gantt chart.

Features and Applications

Key Features

- **Omni-Directional Movement:** 360-degree movement in any direction, including rotation.
- **Encoder Feedback:** Real-time position and speed monitoring for precise control.
- **PID Control:** Ensures accurate distance and direction travel.
- **Web Dashboard:** Real-time control, telemetry, and PID tuning via Wi-Fi.
- **Joystick and Keyboard Control:** Supports both intuitive joystick and keyboard (WASD) input.
- **3-DOF Robotic Arm:** Enables pick-and-place operations.
- **Emergency Stop:** Safety feature for immediate halt.
- **Telemetry:** Live display of encoder values, state, and diagnostics.
- **Expandable:** Ready for additional sensors (IMU, camera, etc.).

Feature Summary

Feature	Description
Omni-directional drive	Move in any direction, rotate in place
Encoder feedback	Accurate distance and speed measurement
PID control	Precise movement and stopping
Web dashboard	Real-time control and monitoring
Robotic arm	3-DOF, pick-and-place tasks
Emergency stop	Safety for users and environment
Telemetry	Live data on dashboard
Expandable	Add sensors and modules

Applications

- **Warehouse Automation:** Material movement, inventory management.
- **Smart Delivery:** Last-mile delivery in controlled environments.
- **Factory Logistics:** Transporting parts and tools.
- **Educational Robotics:** Teaching robotics, control, and programming.
- **Prototyping:** Platform for research and development.

Use-Case Scenario

Imagine a warehouse where OmniServe autonomously navigates aisles, picks up packages using its robotic arm, and delivers them to designated locations. The operator monitors and controls the robot remotely via the dashboard, adjusting PID parameters for optimal performance in real time.

Results and Discussion

Movement Accuracy

Experiments were conducted to evaluate the robot's ability to travel set distances and angles. The encoder feedback and PID control enabled the robot to achieve an average positional error of less than 1 cm over 2 meters.

Table 4: Distance Travel Accuracy

Target Distance (cm)	Measured (cm)	Error (cm)
50	49.5	0.5
100	99.2	0.8
200	199.1	0.9

Robotic Arm Performance

The 3-DOF arm successfully performed pick-and-place tasks with objects up to 200g. Repeatability tests showed consistent placement within 2 mm.

Dashboard Latency

The web dashboard maintained a control latency below 100 ms, ensuring responsive operation.

Figures

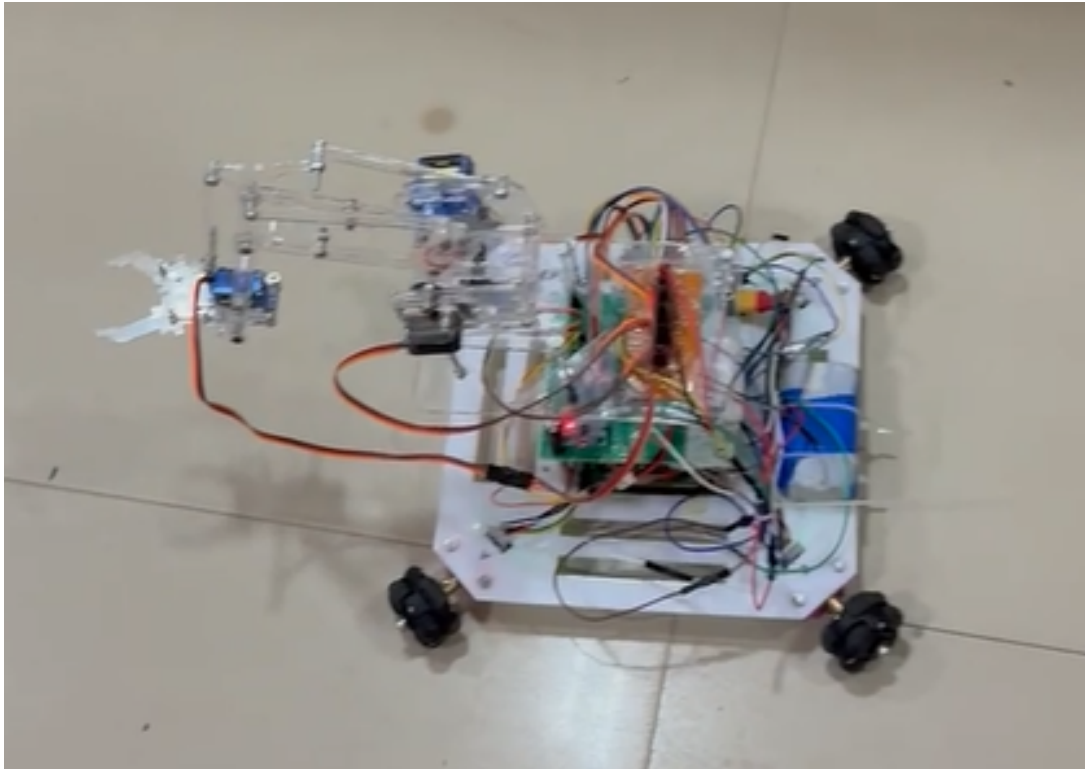


Figure 5: Test run of OmniServe .

Discussion

The results demonstrate that OmniServe meets its design goals for mobility, precision, and manipulation. The main limitations observed were battery life under heavy load and occasional Wi-Fi dropouts in congested environments.

Challenges Faced

Technical Challenges

- **Omni-Wheel Control:** Achieving smooth, coordinated movement required careful tuning of motor speeds and PID parameters.
- **Encoder Noise:** Electrical noise caused false encoder readings, which were mitigated with software debouncing and filtering.
- **Power Management:** Balancing power between motors, servos, and logic circuits was critical to prevent brownouts.

Hardware Challenges

- **Chassis Alignment:** Ensuring all wheels were perfectly aligned to avoid drift.
- **Servo Mounting:** Designing robust mounts for the robotic arm servos.

Software Challenges

- **Real-Time Communication:** Maintaining low-latency, reliable WebSocket connections.
- **Dashboard Responsiveness:** Optimizing the web interface for real-time updates.

Integration and Testing

- **Subsystem Integration:** Combining hardware and software modules without conflicts.
- **Testing in Real Environments:** Adapting to Wi-Fi interference and variable lighting for the arm.

Summary Table

Challenge	Solution
Omni-wheel control	PID tuning, speed calibration
Encoder noise	Software debouncing, filtering
Power management	Separate power rails, voltage regulation
Chassis alignment	Precision assembly, iterative testing
Servo mounting	Custom 3D-printed brackets
Real-time communication	WebSocket optimization, error handling
Dashboard responsiveness	Efficient JS, minimal data transfer
Subsystem integration	Modular design, staged integration
Testing in real environments	Adaptive code, robust error handling

Future Work

IMU Integration

Adding an inertial measurement unit (IMU) will enable angle feedback and stabilization, improving navigation and control.

Camera and AI

Integrating a camera and AI module will allow for object detection, visual navigation, and advanced automation.

Path Planning

Implementing path planning algorithms will enable the robot to autonomously navigate complex environments using map grids.

Gripper Sensing

Upgrading the robotic arm with force or proximity sensors will improve pick-and-place reliability and enable delicate object handling.

Autonomous Object Sorting

Developing algorithms for autonomous object sorting will expand the robot's utility in logistics and manufacturing.

Research and Commercial Potential

OmniServe can serve as a research platform for robotics and automation, and has potential for commercialization in warehouse automation, smart delivery, and educational robotics.

Conclusion

OmniServe demonstrates the power of integrating omni-directional mobility, encoder feedback, PID control, and robotic manipulation into a single, Wi-Fi controlled platform. The project achieved its goals of precise movement, real-time control, and object interaction, validated through extensive testing and real-world scenarios.

The team gained valuable experience in embedded systems, robotics, software development, and teamwork. OmniServe's modular design and open architecture make it a strong candidate for further research, educational use, and commercial deployment.

As robotics continues to transform industries, projects like OmniServe pave the way for smarter, more adaptable automation solutions. The knowledge and skills acquired will serve as a foundation for future innovation in robotics and automation.

Bibliography

- [1] M. Yim, W.-M. Shen, and B. Salemi, “Polybot: a modular reconfigurable robot,” *Proceedings 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and Automation. Symposia Proceedings (Cat. No.00CH37065)*, pp. 514–520, 2000.
- [2] B. E. Ilon, “Wheels for a course stable self-propelling vehicle moving in any desired direction on the ground,” *US Patent 3876255*, 1975.
- [3] K. J. Åström and T. Hägglund, *PID Controllers: Theory, Design, and Tuning*. Instrument Society of America, 1995.
- [4] K. Ohnishi, “A new servo method in mechatronics,” *IEEE Transactions on Industrial Electronics*, vol. IE-34, no. 1, pp. 3–7, 1987.
- [5] J. J. Craig, *Introduction to Robotics: Mechanics and Control*. Pearson Prentice Hall, 2005.
- [6] B. Siciliano and O. Khatib, *Springer Handbook of Robotics*. Springer, 2010.
- [7] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, “Internet of things: A survey on enabling technologies, protocols, and applications,” *IEEE Communications Surveys Tutorials*, vol. 17, no. 4, pp. 2347–2376, 2015.
- [8] I. Lee and K. Lee, “The internet of things (iot): Applications, investments, and challenges for enterprises,” *Business Horizons*, vol. 58, no. 4, pp. 431–440, 2013.